# SCRUM 3

Guitar Shop CRUD

CIS 266

## TEAM MEMBERS & ROLES

- Victoria Mayer – SCRUM Master, code review & polish, DB setup
- Yacob Hiben – Foundational file structure and code, page styles
- Amanda Mays – Coding support

## PROJECT GOALS

- Build a web service that supports CRUD
- Use AJAJ, PHP and a MySQL database
- Create a functional API with testers
- Enable manipulation of data in the guitar shop DB

## APPROACH

**Sprint 1**

- Design and create database
- PHP classes (guitar, customer, order, DB)

**Sprint 2**

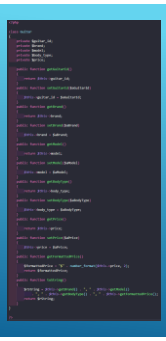
- API endpoint
- API testers

**Sprint 3**

- Client UI and design
- CRUD for all 3 tables

## CLASSES

- Classes Guitar, Customer, and Order use similar structure



## DB CLASS

- Creates connection
- Contains methods to interact with MySQL tables
- Follows pattern using mysqli_query

## API

- Takes request
- Reads data from URL, form post, or JSON
- Puts into array
- Reads table and action

```
$DB_Access = new DB(); // Create an Instance of the DB class

$raw = file_get_contents("php://input");
$body = json_decode($raw, true) ?? array();
$req  = array_merge($_GET, $_POST, $body);
```

```
if ($table == "guitars")
{
    if ($action == "readall")
    {
        $result = $DB_Access->readAllGuitars();
        $data = array();
        while ($row = $result->fetch_assoc()) {
            $data[] = $row;
        }
        print(json_encode($data));
    }
    else if ($action == "read")
    {
        $id = $req['id'];
        $result = $DB_Access->readGuitar($id);
        $data = array();
        if ($result != null) {
            $row = $result->fetch_assoc();
            $data['guitar_id'] = $row['guitar_id'];
            $data['brand']     = $row['brand'];
            $data['model']     = $row['model'];
            $data['body_type'] = $row['body_type'];
            $data['price']     = $row['price'];
        } else {
            $data['error'] = "No record found with guitar_id = $id";
        }
        print(json_encode($data));
    }
    else if ($action == "create")
    {
        $guitar = new Guitar();
```

```
    else if ($action == "update")
    {
        $id = $req['id'];
        $guitar = new Guitar();
        $guitar->setBrand($req['brand']);
        $guitar->setModel($req['model']);
        $guitar->setBodyType($req['body_type']);
        $guitar->setPrice($req['price']);
        $success = $DB_Access->updateGuitar($id, $guitar->getBrand(), $guitar->getModel(), $guitar->getBodyType(), $guitar->getPrice());
        print(json_encode(array("success" => $success ? true : false)));
    }
    else if ($action == "delete")
    {
        $id = $req['id'];
        $success = $DB_Access->deleteGuitar($id);
        print(json_encode(array("success" => $success ? true : false)));
    }
```

## CLIENT AJAX

- HTML with jQuery
- Sends request in background (async)
- Packages JSON object to post to API
- .done writes result

**Guitars - Create**

Brand: Fender

Model: Stratocaster

Body Type: Solid Body

Price: 555

Create

Success: true, New Guitar ID: 13

```
$("#guitars-create-form").submit(function(event) {
    event.preventDefault();
    $.ajax({
        type:  'POST',
        url:  API,
        contentType:  'application/json',
        data:  JSON.stringify({ table: 'guitars',
        action:  'create',
        brand:  $("#guitars-brand").val(),
        model:  $("#guitars-model").val(),
        body_type:  $("#guitars-body_type").val(),
        price:  $("#guitars-price").val() },
        dataType:  'json'
    })
    .done(function(data) {
        $("#guitars-create").html("Success: " + data.success + ", New Guitar ID: " + data.guitar_id);
    });
});
```

## DEBUGGING

- Use built in testers
- View raw API response
- Blank page or errors show problems with PHP
- Check DevTools in the browser

**Provider_3 Testers - Guitar Shop**

All Guitars

1 Gibson Les Paul Solid Body $1580.00
2 Gibson SG Standard Solid Body $1700.00
3 Fender Stratocaster Solid Body $1200.00
4 Fender Telecaster Solid Body $1140.00
5 PRS Custom 24 Solid Body $3620.00
6 Epiphone Casino Hollow Body $999.99
7 Martin D-28 Acoustic $2398.00
13 Fender Stratocaster Solid Body $955.00

**API Tests (JSON)**

GET all guitars
GET all customers
GET all orders


```
JSON   Raw Data   Headers
Save  Copy  Collapse All  Expand All  Filter JSON
▾ 0:
    guitar_id:  "1"
    brand:  "Gibson"
    model:  "Les Paul"
    body_type:  "Solid Body"
    price:  "1580.00"
▾ 1:
    guitar_id:  "2"
    brand:  "Gibson"
    model:  "SG Standard"
    body_type:  "Solid Body"
    price:  "1700.00"
▾ 2:
    guitar_id:  "3"
```


**Customers - Create**

First Name: James

Last Name: Alexander

Email: jalex@email.com

Phone: 480-123-1234

Create

Success: true, New Customer ID: 12